

A pointwise 2-converse: a reader's guide

Companion to OpenAI preprint

OAI:A-pointwise-2-converse-for-elliptic-curves-
with-rational-two-torsion-October-7-2026.

Independent exposition; all mathematics is the preprint's.

October 2026

Status. The preprint [1] is unrefereed, and no Lean or other formalization exists. It is used essentially by OpenAI's preprint on Hilbert's tenth problem over $\mathbb{Q}$ [3]: its §3 needs exactly Theorem 1.1 below to exclude “rank 0 with a divisible part of Sha of 2-corank 1”. Combined with Smith's theorem on the 50/50 distribution of 2^∞ -Selmer coranks in quadratic-twist families [31], Theorem 1.1 would give Goldfeld's conjecture for every $E/\mathbb{Q}$ with rational 2-torsion (twists of such curves keep their rational 2-torsion). Our independent audit (four parallel reviews with numerical checks) found *no error*. It does flag infrastructure that is *sketched* rather than written out, above all the simultaneous ultrafilter-limit diagrams of the ring-class part (Section 8). This guide is an exposition, not a referee report: none of us is a referee in the field, and the result is far beyond the published literature.

How to use this guide. Sections 1–3 (about six pages) give the architecture. Section 4 is the table of $p = 2$ replacements, Section 5 the notation, and Section 6 a walk-through of the paper section by section, with label names and line numbers in the source files. We cite source locations as file:line, for files in `build/sections/`; the numbers are approximate. Paper section numbers: §1 introduction, §2 preliminaries, §3 cyclotomic, §4 ring-limits, §5 ring-determinants, §6 nonvanishing, §7 coefficients, §8 graphs, §9 pointwise. We sometimes simplify notation, and we say so when we do.

A two-hour reading path in the paper.

1. Introduction, introduction:97–160, and the ledger, preliminaries:127–151 (15 min).
2. `cy:binary` and the proof of `cy:missing-vertex`, cyclotomic:121–163, 750–788; then `cy:central`, lines 629–748 (25 min).
3. `rd:det-switch`, ring-determinants:473–565; `rd:binary-transfer` and the proof of `rd:missing-vertex`, lines 1648–1758 (20 min).
4. `pw:odd-lower` and its two inputs, pointwise:355–497; then `pw:odd-base`, lines 872–1027 (25 min).
5. The statement of `gr:address` and the $b = 1$ example, graphs:199–231, 435–454 (10 min).
6. The infrastructure, read against Section 6 below: ring-limits:91–281, 628–777 (25 min).

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1 The theorem, the state of the art, and why $p = 2$ is hard

1.1 Statement

For $A/\mathbb{Q}$ put $r(A) = \text{rank } A(\mathbb{Q})$, $a(A) = \text{ord}_{s=1} L(A, s)$ and

$$s_2(A) = \text{corank}_{\mathbb{Z}_2} \text{Sel}_{2^\infty}(A/\mathbb{Q}),$$

with the local Kummer image at every place. By `pr:ranks` (`preliminaries:13`), $s_2(A) = r(A) + \text{corank}_{\mathbb{Z}_2} \text{Sha}(A/\mathbb{Q})[2^\infty]$.

Theorem 1.1 (`thm:main`, `introduction:22`). *Let $E/\mathbb{Q}$ be an elliptic curve with $E(\mathbb{Q})[2] \neq 0$ and $s_2(E) \in \{0, 1\}$. Then*

$$\text{ord}_{s=1} L(E, s) = \text{rank}_{\mathbb{Z}} E(\mathbb{Q}) = s_2(E), \quad \text{Sha}(E/\mathbb{Q}) \text{ is finite.}$$

In words. A 2^∞ -Selmer corank of 0 or 1 forces the analytic rank to equal that corank. Gross–Zagier–Kolyvagin then gives the rank and finiteness of Sha. The hypothesis is the full Selmer *corank*; it is never replaced by a finite 2-Selmer dimension or by the Mordell–Weil rank.

The theorem covers one rational 2-torsion line or full rational 2-torsion, CM and non-CM curves, every isogeny configuration, and every reduction type at 2 and at the bad primes (`introduction:32–37`). It does not prove the BSD leading-term formula. The broader companion [2] proves a 2-converse without the 2-torsion hypothesis, but is not an input here.

1.2 State of the art

The comparison below follows the paper’s own summary (`introduction:45–76`). We do not add hypotheses beyond what the paper states.

Result	Prime	Range as described in the preprint
Kim [23]	any	“soft” converse: rank = 1 and Sha finite $\Rightarrow a = 1$, every $E/\mathbb{Q}$
Skinner [30]; W. Zhang [35]	odd	rank-one converses at good ordinary primes, under residual and local hypotheses
Castella–Grossi–Lee–Skinner [10, Thm. E]	odd Eisenstein	good reduction, with a local-character restriction
Keller–Yin [22, Thm. B]	odd Eisenstein	potentially good ordinary, without that restriction
Burungale–Tian [7]	every p	CM curves, rank zero
Burungale–Skinner [6, Thms. A.1, A.3]	good ord./ss.	CM, rank one (supersingular with a localization hypothesis)
Burungale–Castella–Skinner–Tian [5]	ordinary	CM, with a Hecke-conductor condition
Kriz [25, Thms. 1.1, 10.17(2)]	ramified in CM field	maximal-order CM, ranks 0, 1; includes the pointwise 2-converse for the congruent-number family
Smith [31]	2	density: 50/50 distribution of s_2 in every twist family; decides no prescribed member
This preprint	2	every $E/\mathbb{Q}$ with $E(\mathbb{Q})[2] \neq 0$, non-CM included, no condition at 2

1.3 Why $p = 2$ with rational 2-torsion is hard

A Kolyvagin- or Kato-style converse at an odd prime p uses, implicitly or explicitly, the following steps. Each one fails or loses control at $p = 2$ when $E[2]$ has a rational point.

- (i) *τ -eigenspace splitting.* Complex conjugation τ splits cohomology into $\pm$ eigenspaces via the idempotents $(1 \pm \tau)/2$. At $p = 2$ these are not integral.
- (ii) *Vanishing of $H^1(\text{Gal}(K(E[2^n])/K), E[2^n])$.* For p odd and large image this gives injectivity of restriction to $K(E[p^n])$ and clean descent of Kolyvagin derivative classes. At 2 it fails in general.
- (iii) *Residual irreducibility.* $E[2]$ is reducible by hypothesis. Classically, irreducibility controls Selmer groups of T/p^n , gives Kolyvagin-system rank bounds, and makes Λ -adic control work.
- (iv) *Real places and Tate cohomology.* Over $\mathbb{Q}$ the real place contributes 2-torsion Tate cohomology. Nekovář’s duality framework assumes his condition (P), “if $p = 2$ then the base field has no real prime”, and treats its absence separately [27, §§5.7, 6.9].
- (v) *Kato’s integral divisibility.* Kato’s integral statement [21, 12.5(4)] (and 13.4(3)) needs $p \neq 2$, according to our audit’s reading of Kato. Only the rational divisibility [21, 12.5(3)] is available at 2.
- (vi) *Small numerical steps.* Examples are the factor $\ell(\ell+1)/2$ in Kolyvagin’s derivative, isotropy of local planes deduced from skew-symmetry, and inertia projectors $(1+g)/2$. Each costs a factor 2 *per prime*, which is fatal when the number of primes is unbounded.

Section 4 lists what replaces each step.

2 The missing-vertex idea

Index quadratic twists by a binary cube. Fix a base h_0 (the curve being tested is $E^{(h_0)}$; in the end $h_0 = 1$). Choose primes $q \in Q$ and linear forms $\lambda_q : \mathbb{F}_2^b \rightarrow \mathbb{F}_2$, and put

$$h_x = h_0 \prod_{q \in Q} (q^*)^{\lambda_q(x)}, \quad x \in \mathbb{F}_2^b, \quad q^* = \left(\frac{-1}{q}\right)q.$$

Address 0 is the unknown curve. The arithmetic constructs one element of a group ring $\mathbb{Z}_2[G]$, $G = (\mathbb{Z}/2)^b$ (with power-series coefficients in practice). Up to a uniformly bounded error, the valuation of its character value at x is the normalized valuation of the L -value, or of the Heegner index, of $E^{(h_x)}$. The core is the following abstract statement, which is our simplified composite of Lemma `cy:binary` (cyclotomic:121) and the proof of Theorem `cy:missing-vertex` (cyclotomic:763).

Lemma A (Abstract missing vertex). *Let $G = (\mathbb{Z}/2)^b$ with characters indexed by $x \in \mathbb{F}_2^b$, and let $u \in \mathbb{Q}_2[G]$ have character values u_x . Suppose*

1. (uniform integrality) $2^C u \in \mathbb{Z}_2[G]$;
2. (bounded valuations off the base) $u_x \neq 0$ and $v_2(u_x) \leq B$ for every $x \neq 0$.

If $M > B + C$ and $b > 2^M - 1$, then $u_0 \neq 0$ and $v_2(u_0) \leq B$.

Proof sketch. Write $z = 2^C u \in \mathbb{Z}_2[G]$. Expanding each sign character as $1 - 2x_i$ gives $z(x) = \sum_I 2^{|I|} a_I \prod_{i \in I} x_i$ with $a_I \in \mathbb{Z}_2$. The j -th binary digit of $z(x)$ is a Boolean polynomial in x of degree at most 2^j . So “ $z(x) \equiv z(0) \pmod{2^M}$ ” is a system of Boolean equations of total degree at most $2^M - 1 < b$. By the characteristic-2 Chevalley–Warning count, the number of common zeros is even; since $x = 0$ is one, there is an $x \neq 0$ with $z_x \equiv z_0 \pmod{2^M}$. As $v_2(z_x) \leq B + C < M$, this forces $v_2(z_0) = v_2(z_x)$. $\square$

With r simultaneous tests the threshold is $b > r(2^M - 1)$ (`cy:binary`). The ring-class version, `rd:bits` and `rd:binary-transfer` (ring-determinants:1620–1699), tests $L(M) + 2$ Laurent coefficients at once.

Why b must be unbounded, and why B must not depend on b . Fourier inversion shows which functions $x \mapsto u_x$ come from $\mathbb{Z}_2[G]$. A simple example (ours, checked by hand) is

$$u = 2^b \cdot 1 - \sum_{g \in G} g, \quad u_0 = 0, \quad u_x = 2^b \ (x \neq 0).$$

Here u is integral, every nonzero vertex has valuation exactly b , and the base vanishes. So a bound of the form $B \geq b$ proves nothing, and for a fixed B the cube must have dimension $b > B$; the lemma asks for $b > 2^{B+C+1} - 1$.

In words. The arithmetic must deliver (1) one integral family whose denominator C does not grow with b , $\#Q$ or the prime count of h_0 , and (2) a valuation bound B at the nonzero vertices that is equally uniform. Only then may b be taken large. Almost every technical device in the paper serves one of these two uniformities at the prime 2.

The two instances.

- *Cyclotomic, rank 0* (`cy:missing-vertex`, cyclotomic:45). Here $u \in \Lambda[G]$ with $2^{C_{\text{int}}} u \in \Lambda[G]$ (`cy:uniform-integrality`), and $|v_2(u_x(0)) - D(h_x)| \leq C_{\text{loc}}$ (`cy:central`), where $D(h) = v_2(\mathcal{L}(h)) - 2w(h) - s(h)$. One applies Lemma A to the constant coefficients.
- *Ring class, rank 1* (`rd:missing-vertex`, ring-limits:64). The coordinates are not uniformly integral. Instead there is a *clearing factor* f with $f_x u_x = d_x$, with $v(f_0(0))$ and the Laurent pole order of f bounded independently of b (`rd:family-clearing`). Lemma A is applied to finitely many Laurent coefficients of f and to the constant coefficient of d .

3 Proof architecture

3.1 Rank zero: the cyclotomic route

If $s_2(E) = 0$, the root number is $+1$ (parity, `pr:invariance`).

1. An *even coefficient system* (**co:even**): a weight-3/2 form whose squarefree coefficients satisfy, within one local squareclass, the exact Waldspurger ratio $2v(a(D)/a(D_0)) = v(L_*(\sigma D)/L_*(\sigma D_0))$.
2. With the cyclotomic lower bound **cy:lower-bound**, this defines a binary *symbol* $A(D) \in \{0, 1\}$: $A(D) = 1$ iff $a(D)$ has the minimal normalized depth $\delta + w(D)$ (**co:minimum**). A unit symbol gives $L_* \neq 0$ and $v(L_*) \leq 2w + C$.
3. The address lemma **gr:address** adjoins prime networks so that the symbol is 1 at every nonzero address of a cube based at $h = 1$.
4. Cyclotomic transfer **cy:missing-vertex** gives $L(E, 1) \neq 0$ (**pw:even-base**, pointwise:306).

The same construction, with a stationary base, gives the uniform *even estimate* of **pw:partitionable-even**, namely $v(L_*(h)) - 2w(h) - s(h) \leq C$ at every “partitionable” h of Selmer corank 0.

3.2 Rank one: three companions

If $s_2(E) = 1$, the root number is -1 . There are three distinct auxiliary discriminants. Keeping them apart is the main bookkeeping point of §§7–9.

$h_* < 0$ (**fixed detector partner**). $L(E^{(h_*)}, 1) \neq 0$. For $h > 0$, the genus-character Heegner sum P_h over $\mathbb{Q}(\sqrt{hh_*})$ has height $\sim L'(E^{(h)}, 1)L(E^{(h_*)}, 1)$, so it *detects* a simple zero of $E^{(h)}$. Its reductions modulo an auxiliary prime ℓ , through the supersingular points of $X_0(N)$, give the *odd coefficients* (**co:theta-bridge**, **co:odd**).

$k'(h) < 0$ (**varying, bounded prime count**). To normalize the odd coefficients one needs the lower bound $2j(P_h) \geq 2w(h) + s(h) - C$, uniform in the prime count of h (**pw:odd-lower**). For each h of analytic rank one, choose k' with $\omega(k') \leq A$ and $L(E^{(hk')}, 1) \neq 0$ (**pw:rough-companion**, using the bounded-factor nonvanishing **nv:rough**). Then combine three things: the ring-class lower bound over $\mathbb{Q}(\sqrt{k'})$ (**rd:ring-lower**, uniform in $\omega(k') \leq A$); the even estimate at hk' ; and the Gross–Zagier height comparison **pw:height-comparison**, $2j_h(k) - 2j(P_h) = v(L_*(hk)) + O(1)$.

$k < 0$ (**fixed final companion**). It is chosen with $L(E^{(k)}, 1) \neq 0$ *before* any cube dimension. Then $s_2(E/K) = s_2(E) + s_2(E^{(k)}) = 1$ for $K = \mathbb{Q}(\sqrt{k})$ (**pw:quadratic-corank**). Cubes based at $h = 1$ have all new primes split in K and two simultaneous unit symbols at each $x \neq 0$: odd at h_x and even at $h_x k$. These give $2j_{h_x}(k) - 4w - s(h_x) - s(h_x k) \leq C$, and the fixed- K ring-class transfer **rd:missing-vertex** returns $\text{ord}_{s=1} L(E, s)L(E^{(k)}, s) = 1$, hence $a(E) = 1$ (**pw:odd-base**).

In words. h_* defines the detector, $k'(h)$ proves its lower bound, and the fixed k permits the final transfer (introduction:138–139).

Order of choices. The ledger at preliminaries:127–151 is worth reading first. Fixed arithmetic constants come first: the curve, the parametrization, the fixed support and local classes. Varying companions may depend only on a bound A for their prime count. The fixed transfer companion K is chosen before b . Then come b , the finite networks, the coefficient precision, and an arithmetic “stage” (a precision M and a reduction prime ℓ). Prime sizes may grow with b ; valuation constants may not.

3.3 Dependency diagram

Figure 1 refines the paper’s Figure 1 (proof-roadmap-figure:1–44). Arrows mean “is used by”.

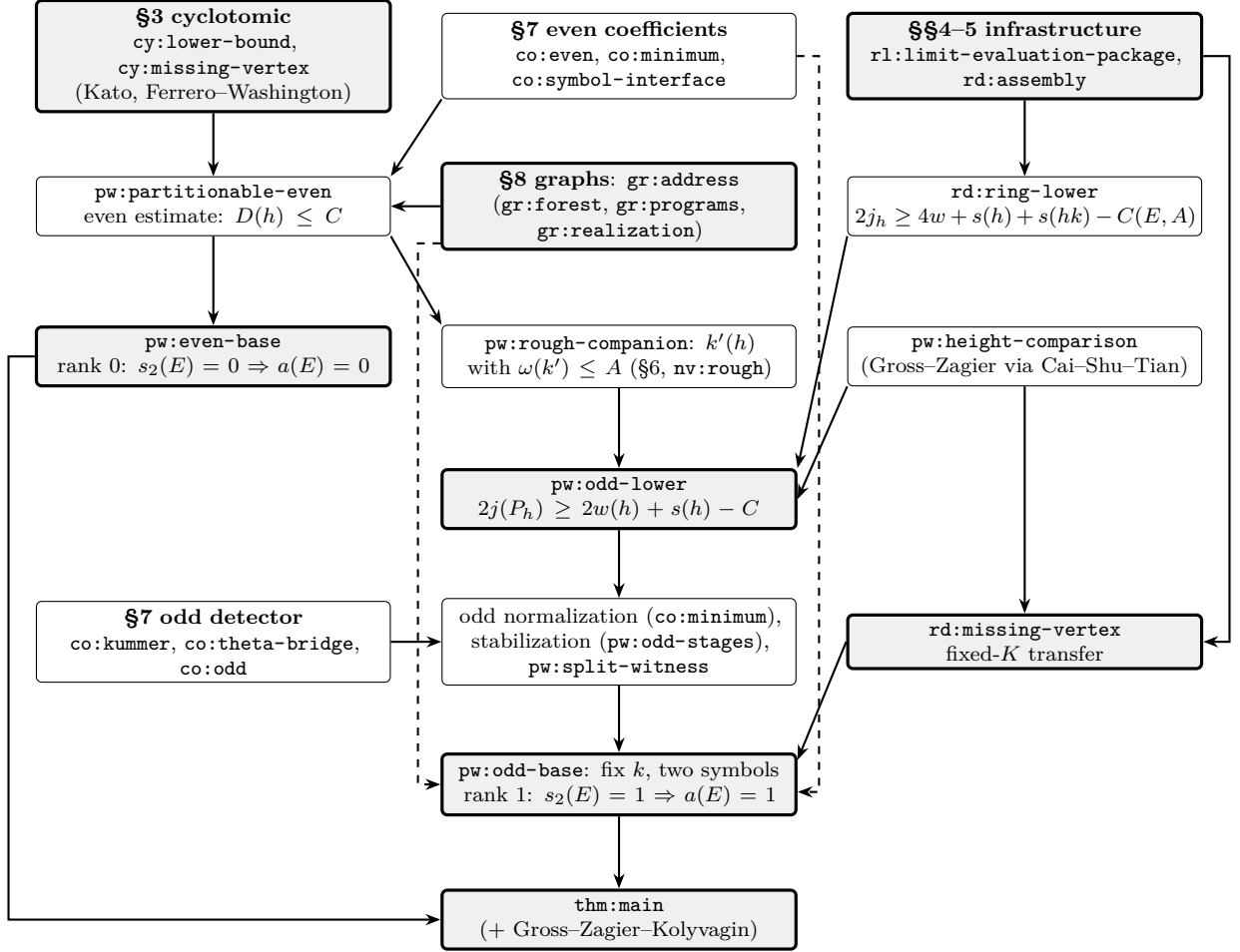


Figure 1: Dependencies among the main statements. Shaded boxes are the main black boxes and the two conclusions. Dashed arrows mark reuse in the final step: the address lemma with two symbols, and the even symbol at hk . The cyclotomic lower bound also normalizes the even coefficients (**co:even-bound**).

4 Classical steps needing p odd, and their replacements

Every entry was checked against the source text. “Sah’s lemma” is our name for the argument; the paper does not use the name.

Classical step	Replacement used here	Where
τ -eigenspaces $(1 \pm \tau)/2$	Work with $(1 + \tau)$ -images. For a crossed cocycle b and an exact-kernel element u , $b((\tau u)^2) = (1 + \tau)b(u)$, and $\text{im}(1 + \tau)$ is a rank-two coefficient module, also in characteristic 2. A rank-two evaluation is selected by polarization of the determinant quadric, which works in characteristic 2. Splitting into signs happens only over characteristic-0 fields.	ring-limits:931–939; rd:rank-two; ring-determinants:666–688
$H^1(\text{Gal}(K(E[2^n])/K), E[2^n]) = 0$	Sah-type central scalar. Fix g acting on T by $u \cdot 1$, $u \in 1 + 4\mathbb{Z}_2$, $u \neq 1$; $z = g^2 \tau g^2 \tau^{-1}$ is ring-class trivial and acts by u^4 ; $(u^4 - 1)b(g) = (g - 1)b(z)$. Restriction to the exact kernel is injective with a loss of fixed valuation. Descent obstruction and ambiguity are killed by a fixed 2^e , and 2^{2e} is paid <i>once</i> for all primes.	rl:kernel ring-limits:731; ring-determinants:235–258; coefficients:220–230
Residual irreducibility (size bounds)	A moving auxiliary character $\Psi = (1 + t)^{\xi_i}$ at split primes r_i . Over $\Omega = \mathbb{F}_2((t))$ the nonconstant scalar makes the local tests acyclic and kills global invariants, so residual Selmer ranks are bounded independently of the active primes.	rl:characters, rl:local-acyclic, rl: residual-bound
Residual irreducibility (rank reduction)	An ultraproduct $\mathcal{O}_\infty = \prod_{\mathcal{U}} \mathcal{O}_j/\mathfrak{a}$, a DVR with uniformizer 2 and <i>characteristic-0</i> fraction field, where T is absolutely irreducible via a finite spanning witness. A hypothetical sequence of unbounded losses becomes a characteristic-0 rank reduction.	rl: outer-evaluation, rl: irreducibility, rd:outer-bound
Residual control, cyclotomic side	Ferrero–Washington $\mu = 0$ bounds class-group 2-ranks in the real cyclotomic tower. Units with odd-augmentation signature (from $\epsilon = 1 + \zeta + \zeta^{-1}$, $N(\epsilon) = -1$) make the signature map surjective. Then $C_S^+ \otimes \mathcal{O}[G] \simeq \mathcal{O}[G]^2[-1]$.	cy:residual cyclotomic:245–311
Real places (Nekovář’s (P))	Ring class: K is imaginary quadratic, so (P) holds and complex places give zero Tate complexes. Cyclotomic: the positive complex $C_S^+ = \text{Cone}(R\Gamma(G_{\mathbb{Q},S}, \mathbb{T}) \rightarrow R\Gamma(\mathbb{R}, \mathbb{T}))[-1]$, an explicit real-place class a , and $2z_S$ lifted because $H^1(\mathbb{R}, \mathbb{T})$ is killed by 2.	ring-limits:182, 242, 399; cy:perfect; cyclotomic:599–609
Kato 12.5(4) (integral)	Kato 12.5(3) (rational, over $\Lambda[1/2]$, characterwise). One fixed modular-form lattice for all traces gives $2^{C_{\text{int}}} u \in \mathcal{O}[G]$, and then $\mathcal{O} \cap \Lambda[1/2] = \Lambda$ (cy:fourier).	cyclotomic:423–438, 597–627
Kolyvagin primes: $p^n \mid \ell + 1, a_\ell$	The derivative reduction is $\frac{\ell(\ell+1)}{2} F_\ell \tilde{X}$, so one requires $v_2(\ell + 1) \geq a + 1$ for vanishing mod 2^a .	ring-determinants:361–378
Isotropy of local planes (skew-symmetry)	Impose $\ell^2 \equiv 1 \pmod{2^{a+1}}$: then -1 is a 2^a -th power and $(x, x) = (x, -1) = 0$ for the 2^a -Hilbert symbol. The pure squares vanish exactly, so the isotropy homotopies can be taken to be zero.	rd:local-model ring-determinants:99–133, 414–419

Classical step	Replacement used here	Where
Primitive localization in each switch	Exact determinant switch over a DVR: $v(u_F) = v(u_S)$ even when the localized class is divisible by 2, so no loss per switch.	<code>rd:det-switch</code> <code>ring-</code> <code>determinants:473</code>
Ordinary/Greenberg conditions at p	The integral Kummer condition $M_F[-1] \rightarrow R\Gamma(F, T_2B)$, including the derived degree-zero torsion, is self-orthogonal at $F \mid 2$ for every reduction type.	<code>rl:kummer</code> <code>ring-limits:285</code>
Smoothing factors ($c^2 - c\sigma_c$)	$c, d \equiv 1 + 2^e \pmod{2^{e+1}}$ make the factor a unit of $\mathcal{O}[G]$ with central valuation exactly e .	<code>cyclotomic:363–389</code>
Inertia projector ($1 + g_q$)/2 at each unused prime	Integral units $U_q = 1 + \frac{r_q-1}{2}(1 + g_q)$ and $\eta = \prod_q (1 + \frac{l_q/e_q^2-1}{2}(1 + g_q))$, using $P_q(Z) - Z^2 P_q(Z^{-1}) = \frac{q-1}{q}(1 - Z^2)$ and $(1 + az + z^2)(1 + az^{-1} + z^{-2}) \equiv (a + z + z^{-1})^2 \pmod{2}$.	<code>cy:</code> <code>euler-convention;</code> <code>rd:unused, rd:eta</code>
Sha over K vs. $\mathbb{Q}$ by $\pm$ -parts	The Cassels–Tate isogeny identity for $\text{Res}_{K/\mathbb{Q}} E^{(h)}$ vs $E^{(h)} \times E^{(hk)}$ (composites = 2) gives a length identity with $\sum_{p k} t_p$.	<code>ring-</code> <code>determinants:1121–1190</code>
Indivisibility / Kolyvagin’s conjecture	No indivisibility is assumed. Binary symbols from half-integral weight coefficients plus prime networks (§8) produce the nonzero vertices with uniform bounds.	<code>co:</code> <code>symbol-interface,</code> <code>gr:address</code>

5 Notation

Symbol	Meaning	Defined
T, V	$T_2E, T \otimes \mathbb{Q}_2$	<code>cyclotomic:6</code>
$s_2(A); s(h)$	Selmer corank; $\text{length}_{\mathbb{Z}_2} \text{Sha}(E^{(h)})[2^\infty]$ (finite)	<code>preliminaries:7, 104</code>
$p^*; h$	$(\frac{-1}{p})p$; squarefree products of $p^*, h = 1$ allowed	<code>preliminaries:88–95</code>
$t_p; w(h)$	$\dim_{\mathbb{F}_2} E[2]^{\text{Fr}_p} \in \{1, 2\}; \frac{1}{2} \sum_{p h} t_p$	<code>preliminaries:97</code>
$L_*(h) = \mathcal{L}(h)$	$\sqrt{ h } L(E^{(h)}, 1) / \Omega_E^{\text{sgn } h}$	<code>cyclotomic:17</code>
$D(h)$	$v_2(\mathcal{L}(h)) - 2w(h) - s(h)$	<code>cyclotomic:27</code>
filter	real sign and unit squareclasses at a support $S \supseteq \{p \mid 2N\}$	<code>preliminaries:80</code>
x, h_x, λ_q	address in $\mathbb{F}_2^b$; cube member; activation form	<code>cyclotomic:50</code>
G, g_q	$(\mathbb{Z}/2)^b$; inertia bit $g_q(x) = (-1)^{\lambda_q(x)}$	<code>ring-</code> <code>determinants:1218</code>
$\Lambda, \mathcal{O}, \Omega$	$\mathbb{Z}_2[[t]]$; 2-adic completion of $\Lambda_{(2)}$; $\mathcal{O}/2 = \mathbb{F}_2((t))$	<code>cyclotomic:67</code>
$C_S^+(\mathbb{T}), a, u$	positive complex; real-place class; $a \wedge \widetilde{2z_S} = u\mathbf{e}$	<code>cyclotomic:188, 196, 607</code>
$C_{\text{int}}, C_{\text{loc}}$	uniform denominator; central comparison constant	<code>cyclotomic:83–88</code>
$K = \mathbb{Q}(\sqrt{k})$	imaginary quadratic, split at $2N$; not $\mathbb{Q}(i), \mathbb{Q}(\sqrt{-3})$ or the CM field	<code>ring-limits:14</code>
P_h (§4)	ring-class sum $\sum_{\sigma} \chi_h(\sigma) \sigma(y_{ h })^*$ over K ; index j_h or $j_h(k)$	<code>ring-limits:28</code>
P_D (§7)	genus-character sum over $K_D = \mathbb{Q}(\sqrt{Dh_*})$, conductor 1; index $j(P_D)$	<code>coefficients:169</code>
$h_*, k'(h), k$	detector partner; varying bounded companion; fixed final companion	§3
A_i, r_i, ξ_i, Ψ	$(\mathbb{Z}/2^i)[t]/(t^i)$; moving split prime; character of order 2^{n_i} ; $(1+t)^{\xi_i}$	<code>ring-limits:111, 407, 536</code>

Symbol	Meaning	Defined
$\ell, f(c), s(c)$	derivative (inert) prime; finite coordinate $c(\gamma^2)$; singular coordinate $c(\sigma)$	ring-determinants:65–79
$\mathcal{D}(C); u_I$	inverse determinant $\bigotimes (\det C^i)^{(-1)^{i+1}}$; coordinate of $[P_I] \otimes [\lambda_I]$	ring-determinants:31, 455
$\mathcal{O}_\infty$	ultraproduct DVR, char. 0 fraction field	ring-limits:790
F, F'	$\mathbb{Q}(E[2]), \mathbb{Q}(E'[2])$ for a 2-isogeny $E \rightarrow E'$	pointwise:129
split/simple	split in F (weight 1) / inert in F (weight $\frac{1}{2}$); <i>good</i> simple if also inert in F'	coefficients:17; pointwise:132
$\delta, A(D)$	minimum normalized depth; binary symbol (1 iff $v(a(D)) = \delta + w(D)$)	co:minimum
B_U, H_U, Θ_J	divided operators, unary-corrected form, unary theta	coefficients:646–697
$t(u), d(u)$	trace and determinant operators on the coefficient lattice	coefficients:786
Δ_z	finite difference $f _{z=0} + f _{z=1}$ in a residue bit	graphs:184

Warning. The letter P_h is used both for the ring-class sum (§4) and, in the introduction and §9, for the genus sum P_D of §7. The indices are written $j_h(k)$ and $j(P_h)$ respectively (**pw:height-comparison** makes the distinction explicit).

6 Section-by-section guide

Headings refer to the *paper's* sections. For each we give (a) the main statements, (b) the idea, (c) the key computation, and (d) what a referee should check.

§2 Preliminaries

preliminaries:1–151

(a) **pr:rank**: $s_2 = r + \text{corank Sha}[2^\infty]$, and analytic rank ≤ 1 gives $r = a$ and Sha finite. **pr:invariance**: $(-1)^{s_2(A)} = \varepsilon(A)$ (Monsky [26]; Dokchitser–Dokchitser [13]); isogeny invariance; and $s_2(A/K) = s_2(A) + s_2(A^{(k)})$, proved with $1 \pm c$ only after rationalization. (b) These fix the corank bookkeeping. (d) Check that the quadratic identity is used only as a corank identity (the paper says so at preliminaries:76). Read the ledger at lines 127–151.

§3 Cyclotomic interpolation

cyclotomic:1–788

(a) **cy:lower-bound**: $v_2(\mathcal{L}(h)) \geq 2w(h) + s(h) - C_E$ whenever $L(E^{(h)}, 1) \neq 0$, uniformly in the number and size of the primes of h . **cy:missing-vertex**: as in Section 2 of this guide, with conclusion $L(E^{(h_0)}, 1) \neq 0$ and $D(h_0) \leq B + C_E$.

(b) *The object.* Work over $R = \Lambda[G]$ with $\mathbb{T} = T \otimes R$, twisted by χ_{h_0} , by the universal quadratic character $\psi : G_{\mathbb{Q}} \rightarrow G$, and by the inverse cyclotomic character. One complex carries all 2^b twists and the cyclotomic variable. After base change to the local ring $\mathcal{O}[G]$, whose residue field is $\mathbb{F}_2((t))$, the positive complex is *free of rank two in degree one* (**cy:residual**). Its determinant is therefore a free rank-one module, and the scalar u defined by $a \wedge \widetilde{2z_S} = ue$ is a single element of $R[1/2]$ whose characters are the twists. The real-place class a is the extra generator that makes the rank 2; it is present because $\mathbb{Q}$ has a real place.

Why two coefficient rings. The ring $\mathcal{O} = \widehat{\Lambda_{(2)}}$ is a DVR with residue field $\mathbb{F}_2((t))$, so $\mathcal{O}[G]$ is local and every element with nonzero image in $\mathbb{F}_2((t))$ is a unit. This is what makes the free model have rank exactly 2, independently of b and of the support. But $\mathcal{O}$ has no evaluation $t = 0$ (cyclotomic:71–75): an element of $\mathcal{O}$ is, modulo 2^M , a Laurent series with finite negative part. Central values must therefore be read in $\Lambda[1/2]$. Lemma **cy:fourier** is the bridge: an element

that is integral over $\mathcal{O}$ and characterwise in $\Lambda[1/2]$ is integral over Λ , with no loss depending on $|G|$.

(c) Key computations.

- *Two-sided integrality*. Kato’s classes (8.1.2–8.1.3), traced through one fixed form lattice and pushed along $[\sigma] \mapsto \chi_{h_0}(\sigma)[\sigma|_{\Gamma_n}][\psi(\sigma)]$ (**cy:coefficient-map**), give $2^{C_{\text{int}}}u \in \mathcal{O}[G]$ with no division by $|G|$. Kato 12.5(3) with Euler corrections gives $u_x \in \Lambda[1/2]$ for each character (**cy:rational-membership**). **cy:fourier** then gives $2^{C_{\text{int}}}u \in \Lambda[G]$.
- *Central formula* (**cy:central**). The integral Poitou–Tate sequence gives length $H^2 = s - i + \sum_q \text{length } H^0(\mathbb{Q}_q, A_h) + O(1)$ (**cy:pt-length**). The index i cancels. Unused good primes give $v_2 \# E^{(h_x)}(\mathbb{F}_q)$, which cancels the imprimitive Euler factors in Kato’s reciprocity (**cy:reciprocity**). Active primes give exactly t_q , summing to $2w(h_x)$.
- *Bad primes*. The correct local block is $H^1(I_q, V_h)$, not $V_h^{I_q}(-1)$ (cyclotomic:486–498; the split multiplicative example is worth reading).

(d) Referee checks: the rational comparison (**cy:rational-comparison**, derived at cyclotomic:523–537), which is *sketched*. The argument is that two rank-one Iwasawa classes agree at infinitely many finite-order characters, and then Weierstrass preparation. Check the uniformity of C_{int} in S, h_0, G, b (cyclotomic:620–622), and the choice of c, d in unsmoothing (Kato 8.1.2 needs $\text{prime}(cd) \cap S = \emptyset$; taking c, d prime suffices). Also check Ferrero–Washington at $p = 2$ for the real cyclotomic tower in **cy:residual**.

§4 Ring-class infrastructure

ring-limits:1–981

(a) Statements of the two outputs, **rd:ring-lower** and **rd:missing-vertex** (lines 42–89). These are followed by the construction lemmas, packaged in **rl:limit-evaluation-package** (line 941).

(b) *What object is being built, and why limits*. We want a perfect Selmer complex over $\Lambda = \mathbb{Z}_2[[t]]$ for $T\chi_h\Psi$, where Ψ is a “ring-class Iwasawa variable” at an auxiliary split prime. Its specialization $t = 0$ should be the Kummer Selmer complex of $E^{(h)}/K$, and its residual fiber over $\mathbb{F}_2((t))$ should have bounded size. (Our gloss, not a sentence of the paper: ring-class extensions of r -power conductor have finite 2-part, and the anticyclotomic $\mathbb{Z}_2$ -extension ramifies above 2, where Ψ is required to be trivial.) The paper therefore uses, at precision i , a *different* prime r_i with a character ξ_i of order $2^{n_i} \rightarrow \infty$ (**rl:characters**), and coefficients $A_i = (\mathbb{Z}/2^i)[t]/(t^i)$. The finite complexes for different i form no inverse system. However, **rl:finite-model** gives bounded free models with explicit contractions (ι_i, p_i, h_i) to the actual cochains, and A_i is finite at each fixed precision. A nonprincipal ultrafilter therefore selects entrywise limits of all matrices, including homotopies and duality data (**rl:finite-limits**).

Two subtleties drive the length of the section. First, a limit cycle need not lift to a genuine finite cycle, and one needs *actual* Galois cocycles in order to evaluate at Frobenius. This is the job of **rl:evaluation**: evaluation matrices $e_{g,i}^1$ are kept for *every* sequence $g = (g_i) \in \prod_i G_K$, with the cocycle identities **rl:eval-boundary-rl:eval-cocycle**. Second, ring-class fields of unbounded degree are never objects of the limit. Descent, Shapiro transfer and coset sums are done at each finite stage, and only target-side data on K are retained (“Target-side data”, **ring-limits:219–281**).

The *outer* limit (**rl:outer-evaluation**) is a compactness device. If losses at (2) were unbounded along a sequence of instances with bounded model size, the ultraproduct DVR $\mathcal{O}_\infty$ (characteristic 0) would carry a limiting complex on which the characteristic-0 rank reduction of §5 runs, and this gives a contradiction.

Schematically (our notation; the top row is the inner limit for one instance, the bottom row

the outer limit over instances j):

$$\begin{array}{ccc} \{(C_i, d_i, \iota_i, p_i, h_i, e_{g,i}^1, \dots)\}_i \text{ over } A_i & \xrightarrow{\lim_{\mathcal{U}}} & C \text{ over } \Lambda \\ & & \downarrow \text{instance } j: \otimes \mathcal{O}_j \\ C_\infty \text{ over } \mathcal{O}_\infty & \xleftarrow{\prod_{\mathcal{U}} / \mathfrak{a}} & \{C_j\}_j \text{ over } \mathcal{O}_j \end{array}$$

All matrices have bounded size. Every asserted identity (chain map, homotopy, duality, cocycle relation) is a finite polynomial identity among their entries, so it passes to the limit. What does *not* pass automatically is the existence of the finite-level arithmetic data. These must be constructed simultaneously at every stage, with one fixed ultrafilter and the same original stages when new derivative primes are added. This is exactly where the audit's Gap 1 lies.

(c) Key points. **rl:kummer**: integral Kummer conditions at 2 keep $B(F)[2^n]$ in derived degree 0. **rl:local-acyclic**: at r , inertia minus one is t -unit, and at active split primes Frobenius is zB with $z = (1+t)^a$, a odd, so it is nonconstant over $\mathbb{F}_2((t))$. **rl:kernel**: the central element $z = g^2 \tau g^2 \tau^{-1}$. **rl:chebotarev-evaluation**: derivative primes with Frobenius τu , evaluating at $(\tau u)^2$; then $a_\ell \rightarrow 0$ and $\ell \rightarrow -1$.

(d) Referee checks: (i) the “Target-side data” paragraph and **rl:limit-evaluation-package** (lines 219–281 and 941–981) are written as checklists pointing to sketched constructions (Audit Gap 1). (ii) In the Kummer orthogonality of **rl:kummer** (lines 320–341), the nullhomotopy is asserted, not exhibited (Gap 2). (iii) Check that one fixed ultrafilter and the same original stages are used after every new switch (ring-limits:206–216; ring-determinants:397–404). (iv) Check the uniform constant c_E in **rl:trace-bound**, including the CM case with $\chi_{\text{cyc}}(\gamma) = 625$.

§5 Switches, determinants, transfer

ring-determinants:1–1769

(a) **rd:derivative**: an integer $L \neq 0$, independent of the number of primes, such that Heegner derivative classes satisfy $f(P_{I \cup \ell}) = 0$ and $s(P_{I \cup \ell}) = Jf(P_I)$. **rd:det-switch**: the valuation is invariant under a switch. **rd:away-two**: the coordinate lies in $\Lambda[1/2]$. **rd:outer-bound**: $v_{(2)}(u) \geq -C(E, A)$. **rd:central-formula**:

$$v(u_h(0)) = 2j_h - s(h) - s(hk) - 4w(h) + O_{E,A}(1).$$

rd:family-clearing: a clearing factor with bounds independent of $b, \#Q$. **rd:missing-vertex** is proved at line 1701.

(b) The determinant coordinate u_I pairs the Heegner class with its own dual functional λ_I , obtained from global duality through local contractions. Hence its central value is $2j_h$ (point and functional each contribute j_h) minus the torsion lengths. A Kolyvagin switch at an inert ℓ replaces the finite plane by the singular plane and preserves $v(u)$ exactly. Repeated switches lower the Selmer dimension of a test fiber, by two at each rank-two switch (a one-sign fiber first needs one preparatory switch), until a 1×1 model makes the coordinate visibly integral. The terminal minor estimate **rd:minor**, $u_i = \pm \alpha_i \beta_i / \det A_i$, then bounds the loss.

(c) Key computations. Kolyvagin's local formulas at 2 (lines 300–378): $f = (a_\ell F_\ell - (\ell + 1))\tilde{Q}$, $s = (a_\ell - (\ell + 1)F_\ell)\tilde{Q}$, and the identity $(F_\ell - a_\ell)(a_\ell F_\ell - (\ell + 1)) = a_\ell - (\ell + 1)F_\ell$, so $s = (F_\ell - a_\ell)f$. The half factor needs $v_2(\ell + 1) \geq a + 1$. In the determinant switch, the vector $v_0 = 2^d e_1$ is paired with the fractional $2^{-d} e_2$: unit volume, with no integral complement needed (lines 553–564). The central trace at r_i is $m_i = a_{r_i} - 2\chi_h(r_i)$, with $v_2(m_i) \leq c_E$. The Sha comparison **rd:sha-comparison** gives length $\text{Sha}(E^{(h)}/K)[2^\infty] = s(h) + s(hk) + \sum_{p|k} t_p + O_E(1)$.

For the family: the universal trace class is built over $A_i[G]$ *before* character evaluation. Unused primes are removed by η , and the new local blocks (residual t -order at most 4) are eliminated by Schur complements with uniform Laurent pole bounds (**rd:schur**). This gives f with $v(f_0(0)) \leq c_*$ and poles $\leq L(M)$.

(d) Referee checks: **rd:assembly** (lines 1542–1604) is the main *sketched* assembly (Gap 1). Its hypotheses “single simultaneous construction” (lines 767–770) are exactly what the infrastructure must deliver; the paper’s own Remark **rd:interface-status** stresses this. Re-derive **rd:det-switch** and the half-factor step yourself (our audit did). Check that Howard is used only as a pattern (lines 407–408, 641): his odd-prime hypotheses are never invoked. The base step uses Gross–Zagier at $h = 1$ via Cai–Shu–Tian (line 1732). The analogous citation at `ring-limits:33` should also point to Cai–Shu–Tian with $c = |h|$ (audit minor note).

§6 Auxiliary twists

nonvanishing:1–425

(a) **nv:local-witness**: in any filter of root number ± 1 there are d with L or $L' \neq 0$, avoiding any finite set. **nv:rough**: in the $+1$ case one can also require $\omega(d) \leq A_0$ with A_0 *absolute*. (b) A first moment in a fixed progression with an imposed divisor b , $S_{0,b}(X) = cX \prod_{p|b} \beta_p + O(b^B X^{1-\eta})$ (**nv:moment**), has nonnegative terms by Waldspurger. A lower linear sieve [15, Thm. 11.13] then gives almost-prime d , following Hoffstein–Luo [18]. (c) Absolute exponents: the Poisson identity **nv:poisson**; explicit local Gauss sums $G_\nu(p^r)$; the bound $(C_0 C_1)^{\omega(J\nu)} \leq (2J|\nu|)^C$. (d) The Poisson and omitted-factor steps are compressed (Gap 2). Check that the factor bound A_0 does not depend on the curve: it is applied to absorbed curves $E^{(h\text{-dummy})}$ in **pw:rough-companion**.

§7 Coefficient systems

coefficients:1–1028

(a) **co:even** (even system and exact ratio); **co:kummer** and **co:theta-bridge**: for ℓ with $\rho_T(\text{Fr}_\ell) \equiv C \pmod{2^M}$, $(h_*/\ell) = -1$, $(D_w/\ell) = 1$ (hence $\ell \equiv -1 \pmod{2^M}$),

$$a_M(D) \equiv c_0 2^{\omega(N)} \zeta_M(D) \lambda_M(\text{red}_\ell(LP_D)) \pmod{2^M}.$$

co:odd: $v(a_M(D)) \geq \min\{M, j(P_D) - C\}$, and a visible coefficient bounds $j(P_D)$. Then **co:minimum**, **co:operators**, **co:traces**, **co:inertia**, and the **co:symbol-interface**.

(b) Even side: in one squareclass, Waldspurger’s comparison is an *equality* of squares, so valuations are exact, with no product of local factors over the new primes. Odd side: a weighted ternary theta series on the definite quaternion algebra ramified at ℓ, ∞ . Its Schwartz function carries local weights: a selected-root character at level primes, ϵ_p on nilpotent orbits at $p \mid h_*$, and a residue-line character at ℓ . Its $|h_*|D$ -coefficients count optimal embeddings weighted by the genus character, which reproduces λ_M of the reduced genus sum. After normalization by the minimum depth δ , coefficients become *bits*. Divided operators B_U and unary-corrected H_U turn Hecke traces at fresh primes into tests of single bits (“fresh-prime testing”).

(c) The contraction identity (**co:contraction**). Changing one oriented ramification bit at $p \in U$ changes the symbol by $A((U \setminus p) \cup q')$ with $\text{Fr}_{q'} \sim g\phi_p$: two prime labels merge, and the *whole* rational prime p is omitted. Two simple vertices give zero. The good-twin rule: inserting two simple primes with equal labels preserves the symbol when $\#E(\mathbb{F}_q)/2$ is odd.

(d) Referee checks: lattice preservation by $t(u), d(u)$ (via Chebotarev approximation by $T_q, q\langle q \rangle$) and the derivation of $H_U[q^2]$ (lines 793–849, 1000–1020) are sketched (Gap 2). Also check the Waldspurger hypotheses (H1), (H2) (lines 93–95), and that the odd normalization is *conditional* until **pw:odd-lower** (lines 566–583).

§8 Arithmetic graphs

graphs:1–646

(a) **gr:realization**: any prescription of independent residue bits involving new vertices is realized by fresh primes (ray classes, Hilbert reciprocity fixes the reverse bits); product clusters can be replaced by single primes. **gr:address** (line 199): given terminals with nonzero terminal polynomials at each $x \neq 0$, for one or two symbols, one can adjoin primary auxiliaries, absent at 0, so that every symbol equals 1 at every $x \neq 0$.

(b) Differentiating in an edge bit contracts that edge (the symbol interface). Trees therefore contract, and cycles differentiate to zero (**gr:forest**). A network of auxiliary vertices acts on the terminal polynomial by a sum of contraction operators. By the all-minors matrix-tree theorem the constant term is $\det Q$ and the pair-contraction term is $c_a^\dagger \operatorname{adj}(Q) c_b$ (**gr:network**), where Q is the grounded Laplacian over $\mathbb{F}_2$.

(c) **gr:programs** builds networks in b groups of three vertices activated by a basis λ_i with $\lambda_i(x_*) = 1$. These are present together only at x_* . The form $H(T, B) = \begin{pmatrix} 0 & T \\ T^\dagger & B \end{pmatrix}$ uses $T = I + P$ (cyclic shift) for the “singular” program, or $T = I + L$ for the pair $\mathcal{N}^{(\varepsilon)}$. For two symbols, the top parts are multiplied in the polynomial ring (not the Boolean ring) and the sum over ε is shown to be 1 (lines 559–625).

(d) Our audit brute-forced the combinatorics (400 random graphs, about 200 full runs of **gr:address**, 0 failures). The referee should focus on the arithmetic input: that the symbols really satisfy whole-prime omission and primary isolation (§7 and §§9.5–9.6). Note the closing caveat (lines 637–646): this section proves no uniformity in b .

§9 Assembly

pointwise:1–1058

(a) **pw:interfaces**; **pw:partitionable-even**; **pw:even-base**; **pw:rough-companion**; **pw:height-comparison**; **pw:odd-lower**; **pw:split-witness**; **pw:odd-base**; the proof of **thm:main** (line 1029).

(b) Isogeny orientation: use a full-2-torsion curve in the isogeny class if there is one; otherwise F, F' are quadratic and “good simple” primes have $\#E(\mathbb{F}_p) \equiv 2 \pmod{4}$ (**pw:good-simple**). Seeds, reserves and neutral packets (lines 149–274) are finite label combinatorics. They show that a long list of terminal primes can be contracted to a minimum seed plus good twin pairs, so the terminal polynomial is nonzero. One packet is put on every nonzero linear form of $\mathbb{F}_2^b$. The odd symbol is stabilized along an ultrafilter of precisions M (lines 499–549); an actual stage is chosen only after b and the networks.

(c) The full-2-torsion case needs an extra argument (**pw:split-witness**). After restricting to K -split primes the even minimum might sit only at the empty configuration. The paper shows $a_p(E^{(k)}) \equiv p+1 \pmod{8}$ for split p , divides once more by 2 (**pw:four-divided-form**), and finds a nonempty witness from a prime inert in K and trivial on $E[8]$. The orientation correction **pw:coinvariants** uses $\operatorname{ann}((V_0^*)^c) = (1+c)V_0$.

(d) Check the order of choices in **pw:odd-base**: k is fixed before b , and ℓ eventually avoids k . Check the final inequality $2j_h(k) \leq 4w(h) + C$ (line 1017). The audit notes that **pw:split-witness** should state that k is a square at $2N$.

7 Worked examples

7.1 A cube for $E = 14a1$

Take $E : y^2 + xy + y = x^3 + 4x - 6$, conductor 14, $E(\mathbb{Q})_{\text{tors}} \cong \mathbb{Z}/6$, so $E(\mathbb{Q})[2] = \mathbb{Z}/2$. Its discriminant is $-21952 = -2^6 \cdot 7^3$, so $F = \mathbb{Q}(E[2]) = \mathbb{Q}(\sqrt{-7})$ and $t_q = 2$ iff $(\frac{-7}{q}) = 1$. With PARI (computation ours) we get $L(E, 1)/\Omega_E^+ = 1/6$, so $v_2(\mathcal{L}(1)) = -1$; this matches BSD with $c_2 c_7 = 6$ and $\#E(\mathbb{Q})_{\text{tors}} = 6$.

The filter of $h_0 = 1$: $d > 0$, $d \equiv 1 \pmod{8}$, $(\frac{d}{7}) = 1$. A cube with $b = 2$ uses $\lambda_{-3} = \lambda_{-19} = x_1$ and $\lambda_5 = \lambda_{13} = x_2$:

$$h_{00} = 1, \quad h_{10} = 57 = (-3)(-19), \quad h_{01} = 65 = 5 \cdot 13, \quad h_{11} = 3705.$$

All four lie in the filter.

What to expect (heuristic, assuming BSD; not used anywhere). For a rank-0 twist, BSD writes $\mathcal{L}(h)$ as $\#\text{Sha} \cdot \prod_p c_p$ over torsion, times period comparisons. At an active prime $p \mid h$ the twist has additive reduction, and the 2-part of c_p is 2^{t_p} (the paper uses this at ring-determinants:1170–1178). Up to bounded local factors at $2N$, one expects $v_2(\mathcal{L}(h)) = 2w(h) + s(h) + O(1)$, i.e. $D(h) = O(1)$. Theorem `cy:lower-bound` proves the lower half of this without BSD. The table lists $\mathcal{L}(d) = \sqrt{d} L(E^{(d)}, 1) / \Omega_E^+$ and $s_{\text{an}} = v_2$ of the BSD-predicted order of Sha (analytic; it is *not* a theorem input).

d	factors	$\mathcal{L}(d)$	$v_2(\mathcal{L})$	$2w(d)$	s_{an}	$D(d)$
1	—	1/6	−1	0	0	−1
57	(−3)(−19)	6	1	2	0	−1
65	5 · 13	6	1	2	0	−1
3705	(−3) · 5 · 13 · (−19)	24	3	4	0	−1
305	5 · 61	54	1	2	0	−1
1241	17 · 73	24	3	2	2	−1
1649	17 · 97	96	5	2	4	−1
1817	(−23)(−79)	96	5	4	2	−1
2937	(−3)(−11) · 89	96	5	4	2	−1
473	(−11)(−43)	0	—	4	—	—

In the filter, among the 46 products $d \leq 4000$ of at most three signed primes below 100 (other than 2, 7; the empty product included), 36 have $\mathcal{L}(d) \neq 0$, and all 36 have $D(d) = -1$ (with $s = s_{\text{an}}$). This is consistent with `cy:lower-bound` and with the shape of `cy:central`, where all varying contributions to $v_2(\mathcal{L})$ are $2w + s$. Without s , $v_2(\mathcal{L}) - 2w$ takes the values $-1, 1, 3$, as in the audit’s 138-twist check. *In words:* the active primes contribute exactly t_q each, and Sha the rest, up to a constant. The uniform bound $B = -1$ at the three nonzero vertices of the cube is the kind of input `cy:missing-vertex` needs. (Here the base is known anyway; the theorem needs arbitrarily large b , built by §8.)

7.2 The smallest address network

The paper’s concrete instance (graphs:435–454): $b = 1$, target $x_* = 1$, two always-active split terminals a_0, a_1 with selected bit z , and three primary vertices 1, 2, 3 activated by x . The program gives the path

$$a_0 - 1 - a_1 - 3 - 2, \quad Q = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \quad c_{a_0} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad c_{a_1} = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}.$$

We checked by enumeration: (i) Q is the grounded Laplacian over $\mathbb{F}_2$ of this path; (ii) the forests with exactly one terminal per component are the two obtained by deleting $a_0 - 1$ or $1 - a_1$, so $\det Q = 2 \equiv 0$; (iii) the only forest joining a_0, a_1 with all vertices covered is the whole path, and $\text{adj}(Q) = e_1 e_1^t$, so $c_{a_0}^t \text{adj}(Q) c_{a_1} = 1$. *In words:* at $x = 1$ the network acts on the terminal polynomial exactly by Δ_z ; at $x = 0$ it is absent and acts by the identity. If the symbol’s polynomial is z (a bit that must be flipped), the network flips it at $x = 1$ only.

8 Audit status

What was checked. Four independent reviews: cyclotomic; ring class; nonvanishing, coefficients and graphs; assembly and sanity. Numerical checks used `cypari2` and `numpy`:

- root numbers and rank bounds on 446 curves $y^2 = x^3 + ax^2 + bx$;

- 138 twists of 14a1, with $v_2(L_*) - 2w \in \{-1, 1, 3\}$;
- Tamagawa factors in 302 cases;
- the graph combinatorics (400 random graphs, about 200 full runs of `gr:address`, 0 failures).

Each cited result was checked against its $p = 2$ hypotheses: Kato 12.5(3) and the integral classes 8.1.3, 8.9–8.12 and 9.7 hold at $p = 2$, while 12.5(4) and 13.4(3) are avoided. Ferrero–Washington holds at 2, and Nekovář’s (P) holds over imaginary K . Howard is used only as a pattern. **Verdict:** **no error found.**

Gaps (sketched or unverified, not shown false).

1. *Simultaneous-diagram infrastructure*: ring-determinants:1542–1604, ring-limits:219–281, 941–981. Ultrafilter limits of finite-level Selmer complexes with moving auxiliary and derivative primes are written as checklists pointing to sketched constructions. This is the place most in need of expert scrutiny.
2. *Smaller sketches*: the rational comparison cyclotomic:523–537; the Kummer orthogonality nullhomotopy ring-limits:320–341; lattice preservation by $t(u), d(u)$ and the derivation of $H_U[q^2]$ (coefficients); the compressed Poisson and omitted-factor steps (nonvanishing).
3. *Not independently verified*: Kato’s 13.4 rests on other works at $p = 2$, and these were not rechecked.

Minor fixes. The key Howard2012 is Howard’s 2004 Compositio paper. ring-limits:33 should cite Cai–Shu–Tian with $c = |h|$. Kato 8.1.2 needs $\text{prime}(cd) \cap S = \emptyset$ (take c, d prime). The real-component cokernel (Kato 12.1.2) is unmentioned but harmless. pw:split-witness should state that k is a square at $2N$. P_h denotes two different sums.

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